2026 Illinois Stats Datathon

Team 030 – Databricks Aficionados

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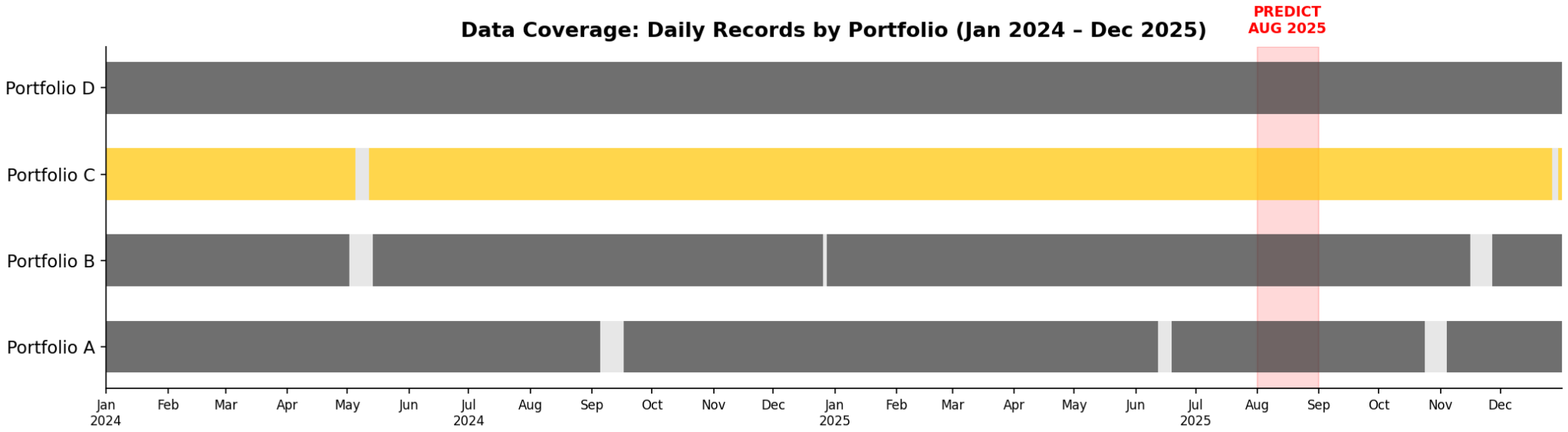
Objective

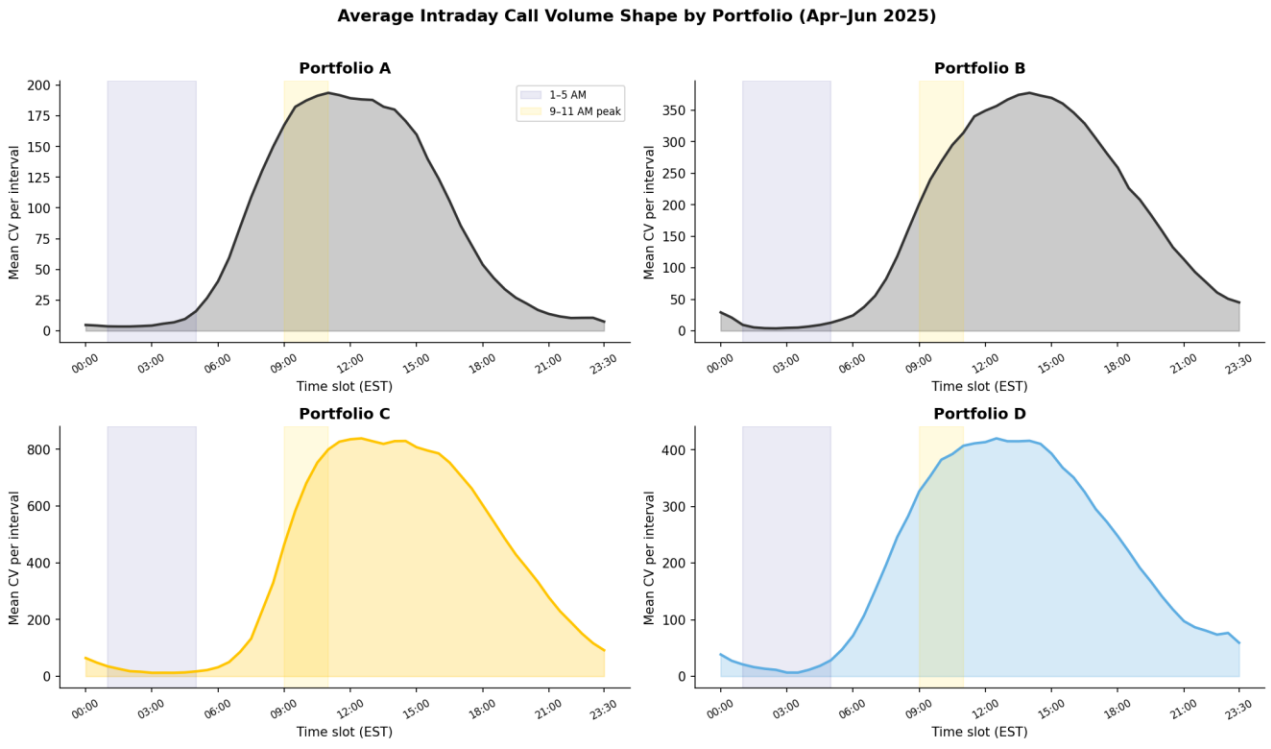
- Forecast Inbound Call Volume for August 2025
- Predict Customer Care Time and Abandon Rate
- Optimize Staffing Decisions

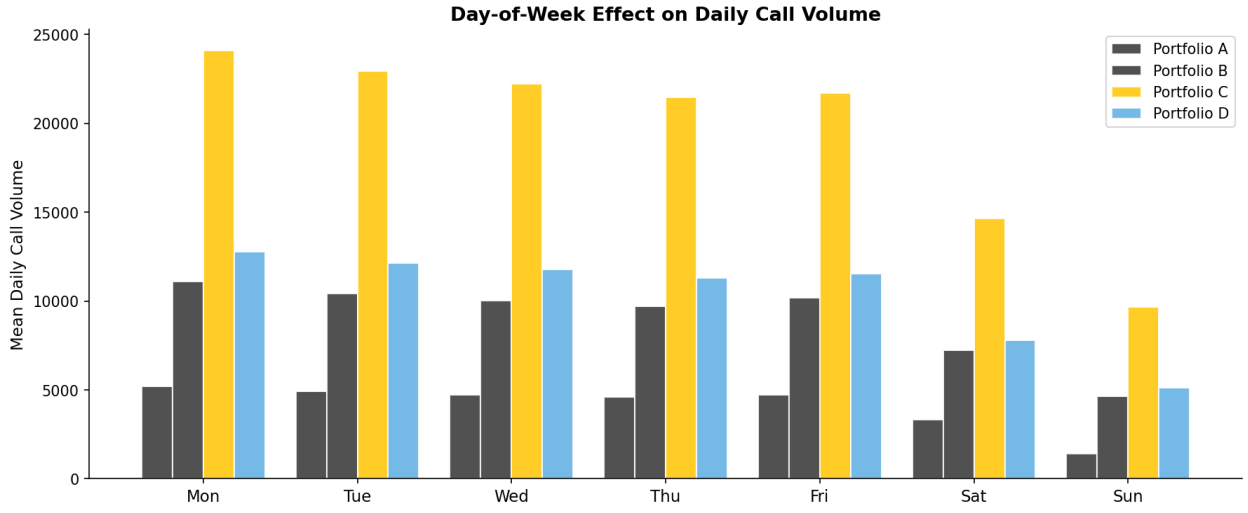
Because understaffing is more expensive than overstaffing, our design deliberately leans conservative.

Data Overview

- **Daily Data**: Jan 2024 – Dec 2025 | Portfolios: CV, CCT, ABD | 701–731 rows per Portfolio
- **Interval Data**: Apr – Jun 2025 | 48 slots/day (30-min intervals) | 3,911–4,313 rows per Portfolio
- **Staffing Data**: Jan – Dec 2025 | 365 rows | Agents per day per Portfolio
- **Prediction Target**: August 2025 *(present in daily data, held out from training)*
- **Missing Daily Days**: A: 30 | B: 25 | C: 10 | D: 0







The Intraday Shape

Intraday Shape *(consistent across all 4 portfolios; scale differs, shape does not)*

- **Peak zone**: 9–11 AM EST
- **Overnight dead zone**: 1–5 AM EST — near-zero volume across all Portfolios

Day-of-Week Effect

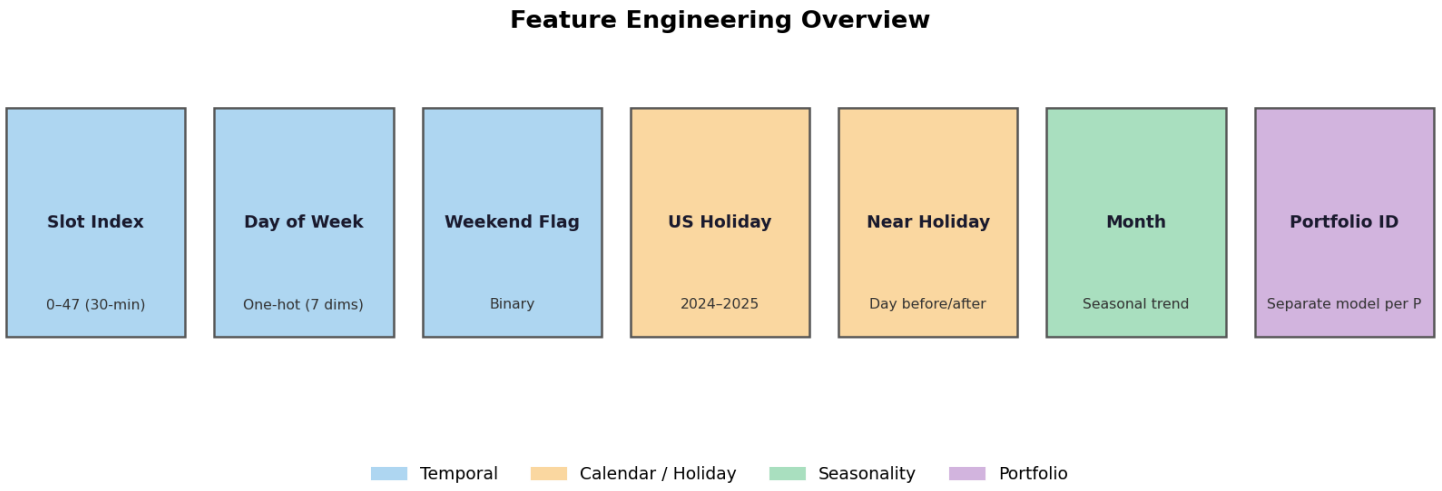
- Monday = highest volume | Sunday = lowest volume
- Magnitude of swing varies by Portfolio

Monthly Trend

- Longer-term seasonality visible in monthly trend chart

Feature Engineering

- **Temporal**: Slot index (0–47), day of week (one-hot, 7 dims), weekend flag
- **Calendar &** Holidays: US holiday flag (2024–2025), near-holiday flag (day before/after)
- **Seasonality**: Month encoding for seasonal trend capture
- **Portfolio Modeling**: Separate model per Portfolio *(no shared parameters)*



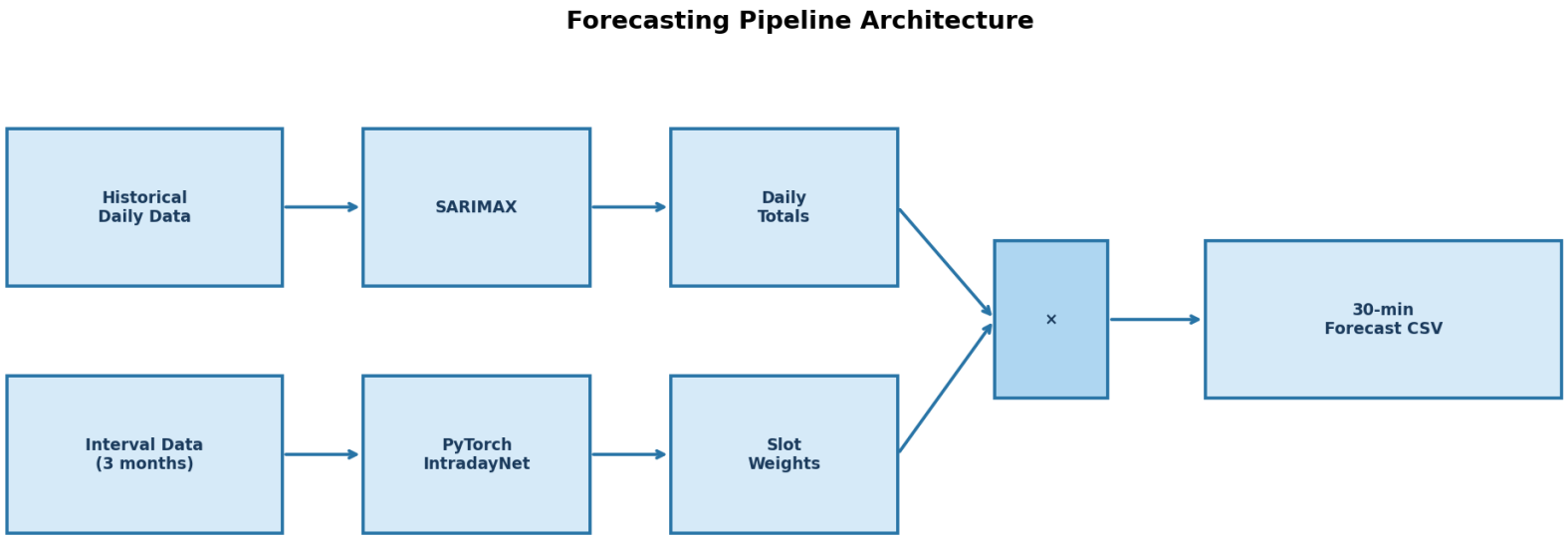
Feature Routing

- Calendar features → SARIMAX exogenous regressors
- Day-of-week → PyTorch intraday network

Near-holiday captures the day-before surge common in financial services, where customers call ahead of billing deadlines.

Model Architecture

- **Stage 1 — Daily Totals (SARIMAX)**
- SARIMAX(1,1,1)(1,1,1,7) per Portfolio per metric — 12 models total *(4 Portfolios × 3 metrics)*
- **Exogenous regressors**: IsWeekend, IsHoliday, NearHoliday
- **Output**: 31 daily forecasts for August 2025 (CV, CCT, ABD)



- **Stage 2: Intraday Distribution (PyTorch)**
  - **Architecture**: Linear(7→64) → ReLU → Linear(64→48) → Softmax
  - **Input**: day-of-week one-hot (7 dims) → Output: 48 weights summing to 1.0
  - **Trained per Portfolio**: 90 days of interval data, 300 epochs, Adam + weight decay 1e-4
  - **4 IntradayNet models** (1 per Portfolio) used for CV distribution
  - **CCT & ABD** Intraday: Empirical slot-level means *(more stable than neural net for these metrics)*

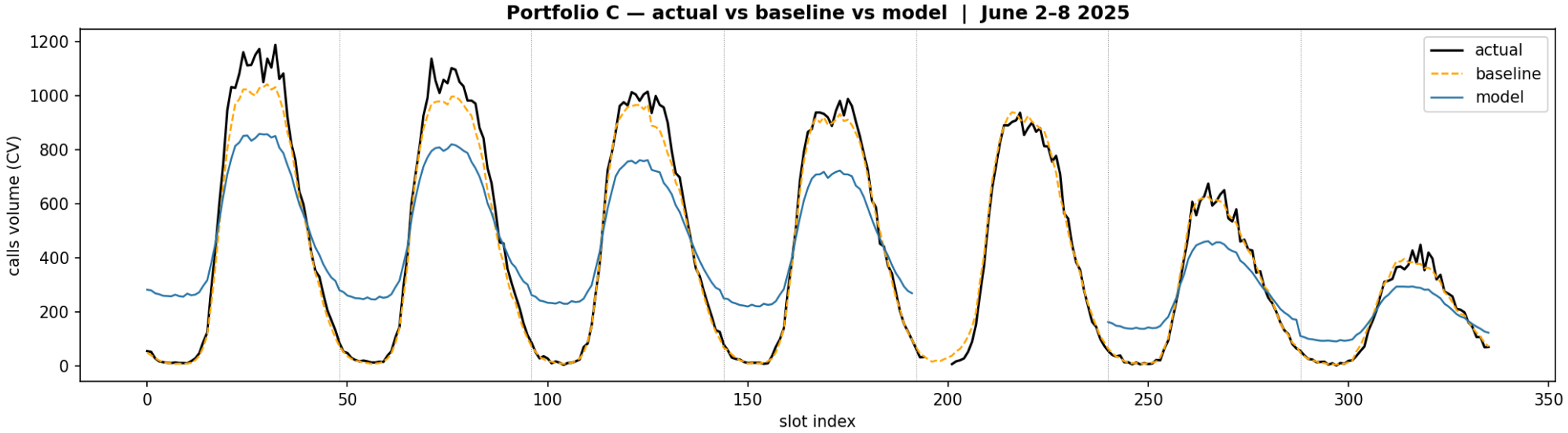
Why This Beats Baseline

Validation Approach

- **Baseline**: Naive empirical mean per (day-of-week, slot) from Apr–Jun interval data
- **Chart**: Portfolio C, June 2–8 | Actual (black) vs Baseline (orange) vs Model (blue)
- **MAPE Note**: Interval-level MAPE inflated by near-zero overnight slots (1–5 AM); denominator instability drives 700%+ errors on low-volume intervals, not a signal of model quality

Why the Model Outperforms Baseline

- Baseline captures the average intraday shape but cannot adapt to day-specific volume variation
- Model combines SARIMAX daily totals × empirical intraday weights → scales correctly to each day's actual demand level
- Monday spikes & weekend dips: baseline over-smooths, model follows — visible in the line chart above
- **Optimization target is composite penalty score (understaffing cost > overstaffing cost), not MAPE**: a deliberate design choice given the asymmetric scoring function



Results

Submission Output — 1,488 rows | 31 days × 48 slots × 3 metrics × 4 Portfolios

Daily Forecast Charts

- **CV**: Weekly seasonality clearly visible: Monday peaks, weekend dips across all Portfolios
- **CCT**: Mean handle time stable within each Portfolio, varies across Portfolios
- **ABD**: Low overall abandon rate; spikes align with high-volume days

  Intraday Preview: **Monday Aug 4** *(4-panel, one per Portfolio)*

- **Learned 30-min shape**: morning ramp → afternoon peak → overnight near-zero

Output Integrity

- All values non-negative, ABD capped to **[0, 1]**, CV includes **+7%** upward bias

The Asymmetry Decision

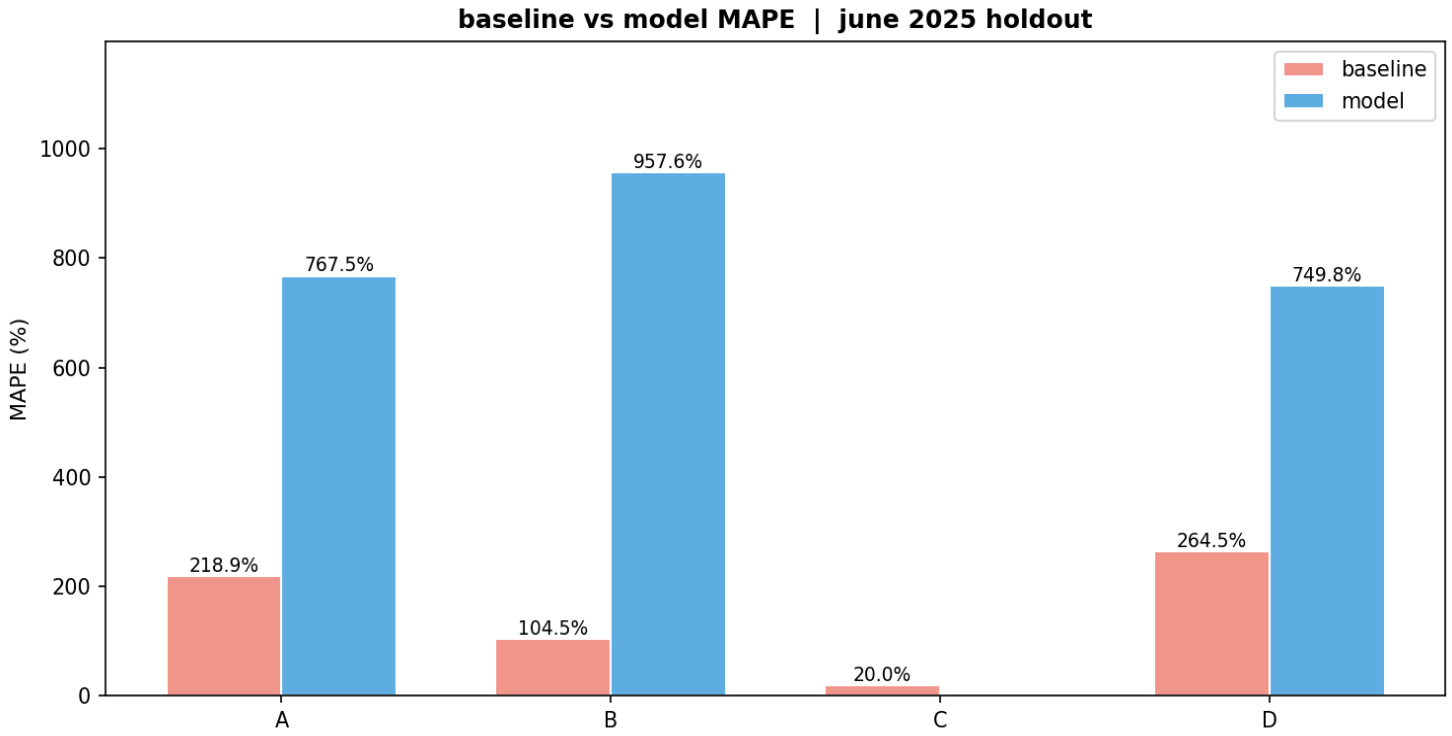
Asymmetric Cost & Bias Strategy

- **Scoring Formula**: Penalty Workload = α × understaffing + β × overstaffing, where α > β
- **Our Decision**: Apply +7% upward bias to all CV forecasts post-prediction
- **Composite Score** = EV + EC + EB + Penalty Workload *(we optimize total score, not MAPE alone)*

Effect of Upward Bias on Error Distribution

- ↓ Fewer intervals where Forecast < Actual *(understaffing — expensive)*
- ↑ More intervals where Forecast > Actual *(overstaffing — cheaper penalty)*

Scope: Bias applied to CV only; CCT & ABD use empirical means *(staffing is driven by call volume)*



Business Impact

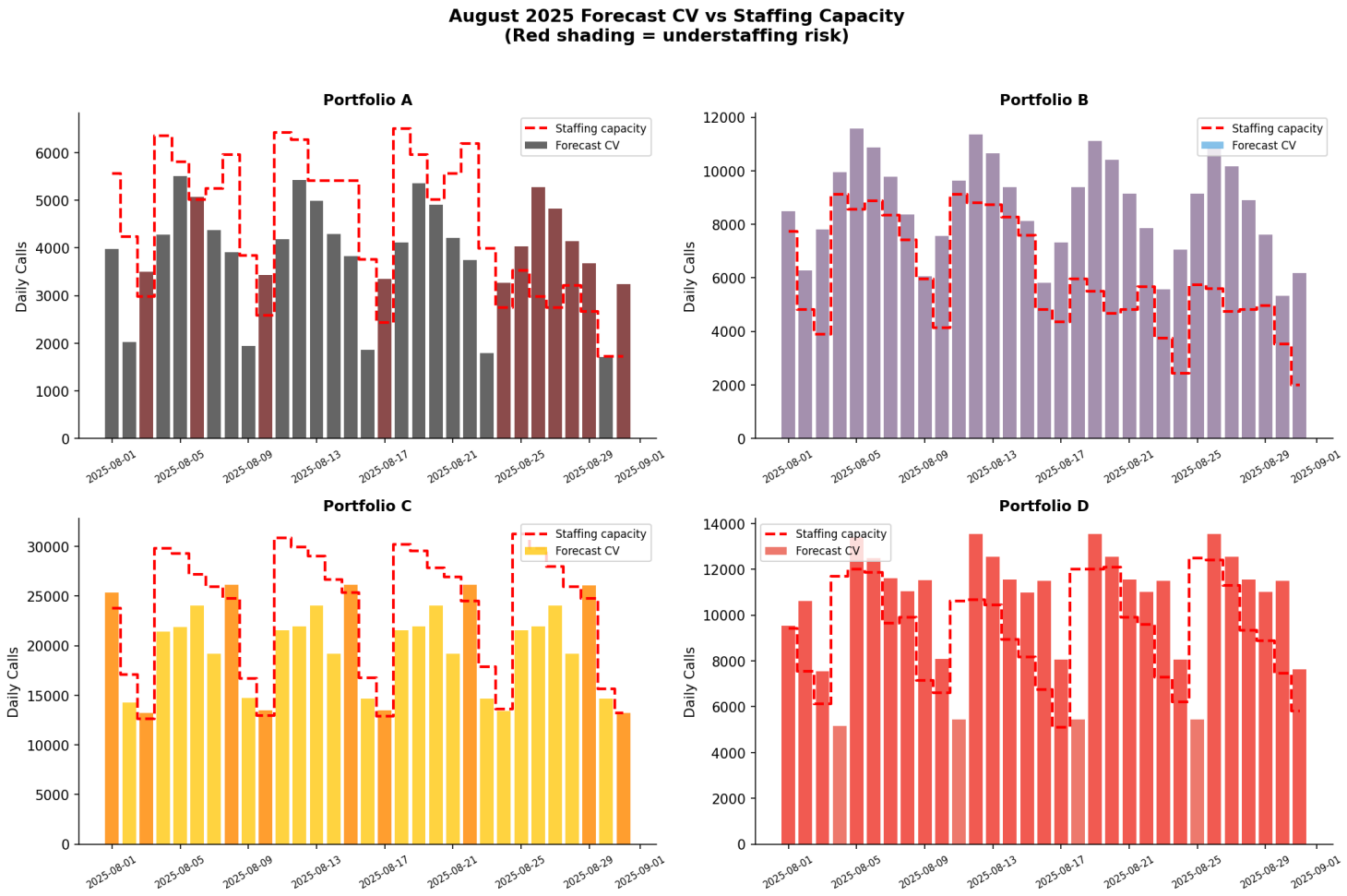
Forecast Output & Staffing Integration

- **Capacity Formula** — Agents × 85% utilization × 8h shift ÷ avg CCT per interval
- **Total Forecast Points** — 5,952 *(4 Portfolios × 31 days × 48 intervals)*
- **Submission Output** — 1,488-row CSV matching template *(31 days × 48 slots)*

  4-Panel Staffing Chart *(one panel per Portfolio)*
- Daily CV forecast bars vs. staffing capacity line
- Red-shaded bars = understaffing risk days *(forecast CV exceeds capacity)*

Deliverable

Not just a CSV — an early-warning system for SLA risk days



Future Work

Deep Sequence Models

- LSTM / Temporal Fusion Transformer once interval history exceeds 3 months
- Current 90-day window is insufficient for sequence architectures

External Signals

- Incorporate billing cycle dates, promotions, and marketing events
- These drive predictable volume spikes not visible in historical call data

Cross-Portfolio Learning

- Shared infrastructure means correlated volume across Portfolios
- Especially valuable for Portfolio A *(30 missing days, noisiest signal)*

Automated Retraining Pipeline

- Daily forecast vs. actuals monitoring → weekly retraining
- Tune bias coefficient against realized penalty scores
- Transforms a one-shot prediction into an adaptive, self-improving system

Thank You!

Solution Summary

Architecture

- SARIMAX (daily totals) + PyTorch IntradayNet (intraday distribution) → 30-min forecasts
- 12 SARIMAX models + 4 IntradayNet models = 16 trained models total

Output

- 1,488 rows × 19 columns matching submission template exactly
- CV, CCT, ABD at 30-min intervals — all 31 August days × 4 clients

Design Principles

- **Asymmetric**: +7% CV bias creates a deliberate overstaffing lean to minimize penalty score
- **Explainable**: 2 stages with clear, distinct purposes: daily scale and intraday shape
- **Repeatable**: Single notebook sequence: 01 → 02 → 03 → 04
- **Scalable**: Separate models handle 5x volume difference (Client A vs C) without compromise